

→ Expression?

An expression is a combination of constants, variables and operations.
operations $\rightarrow (+, -, \times, \div)$

exr. $a + 5$
or $4 - x^2$

Equation:- An equation is a combination of two expressions with equal sign.

exr. $5x + 2 = 0$

→ polynomials:-

An algebraic expressions in which the exponent is a positive integer or zero is called polynomial expression or polynomial.

or:-
An algebraic expression in which variable having powers only from set of whole number is called polynomials.

ex:- positive integers:

$$\mathbb{Z} = \{0, +1, +2, +3, \dots\}$$

$$\omega = \{0, 1, 2, 3, 4, 5, \dots\}$$

ex/ $7n^1 - 4n^0$

is a polynomial expression

ex/ $\frac{\sqrt{n} + 1}{n} = \frac{n^{\frac{1}{2}} + 1}{n}$ non-polynomial

9th class Small Board Mathematics

⇒ Examine whether the following algebraic expressions are polynomial or not?

ex: i -1 ✓

so here there is no variable
Hence it is polynomial expression.

Ans:

ii $\frac{3}{y}$

variable is in $3y^{-1}$
means its power is in the
negative form, hence
not a polynomial expression

iii $\frac{1}{n} - n$

~~$\frac{1}{n}$~~ $n^{-1} - n$

Hence it is also not a polynomial

→ for polynomial:

Notes: variable power (-ve)

should not included.

means whatever the
expression is given to us

we have to see just variable's

power, that should not
be negative.

if Any then not a
polynomial.

Note: 2

variable's power should
not be in fraction like

$\frac{1}{y}$ form

iv, $\sqrt{5} n + y$

$5^{-\frac{1}{2}}$ is the polynomial expression.
 (it is constant its power could be in fraction form.)

v, $5ny^3$

yes it is polynomial.

vi, $\sqrt{5n}$
 $5(n)^{\frac{1}{2}}$

not a polynomial expression.

ix, $\frac{1}{n^2 + 3}$

$1n^{-2} + 3^{-1}$ Not polynomial.